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Baxter

PeriOLIMEL N4E

Emulsion for infusion

1. NAME OF THE MEDICINAL PRODUCT

PeriOLIMEL N4E, emulsion for infusion.

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

PeriOLIMEL N4E is presented in the form of a 3-compartment bag. Each bag contains a glucose solution with calcium, a lipid emulsion and an amino acid solution with other electrolytes.

	Content per bag		
	1000 ml	1500 ml	2000 ml
18.75% Glucose solution (corresponding to 18.75 g/100 ml)	400 ml	600 ml	800 ml
6.3% Amino acid solution (corresponding to 6.3 g/100 ml)	400 ml	600 ml	800 ml
15% Lipid emulsion (corresponding to 15 g/100 ml)	200 ml	300 ml	400 ml

Composition of the reconstituted emulsion after mixing the content of the 3 compartments:

Active substances	1000 ml	1500 ml	2000 ml
Refined olive oil + refined soybean oil ^a	30.00 g	45.00 g	60.00 g
Alanine	3.66 g	5.50 g	7.33 g
Arginine	2.48 g	3.72 g	4.96 g
Aspartic acid	0.73 g	1.10 g	1.46 g
Glutamic acid	1.26 g	1.90 g	2.53 g
Glycine	1.76 g	2.63 g	3.51 g
Histidine	1.51 g	2.26 g	3.02 g
Isoleucine	1.26 g	1.90 g	2.53 g
Leucine	1.76 g	2.63 g	3.51 g
Lysine (equivalent to Lysine acetate)	1.99 g (2.81 g)	2.99 g (4.21 g)	3.98 g (5.62 g)
Methionine	1.26 g	1.90 g	2.53 g
Phenylalanine	1.76 g	2.63 g	3.51 g
Proline	1.51 g	2.26 g	3.02 g
Serine	1.00 g	1.50 g	2.00 g
Threonine	1.26 g	1.90 g	2.53 g
Tryptophan	0.42 g	0.64 g	0.85 g
Tyrosine	0.06 g	0.10 g	0.13 g
Valine	1.62 g	2.43 g	3.24 g
Sodium acetate, trihydrate	1.16 g	1.73 g	2.31 g
Sodium glycerophosphate, hydrated	1.91 g	2.87 g	3.82 g
Potassium chloride	1.19 g	1.79 g	2.38 g
Magnesium chloride, hexahydrate	0.45 g	0.67 g	0.90 g
Calcium chloride, dihydrate	0.30 g	0.44 g	0.59 g
Glucose anhydrous (equivalent to Glucose monohydrate)	75.00 g (82.50 g)	112.50 g (123.75 g)	150.00 g (165.00 g)

^a Mixture of refined olive oil (approximately 80%) and refined soybean oil (approximately 20%) corresponding to a ratio essential fatty acids / total fatty acids of 20%.

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For a full list of excipients, see section 6.1.

Nutritional intakes of reconstituted emulsion for each of the bag sizes:

	1000 ml	1500 ml	2000 ml
Lipids	30 g	45 g	60 g
Amino acids	25.3 g	38.0 g	50.6 g
Nitrogen	4.0 g	6.0 g	8.0 g
Glucose	75.0 g	112.5 g	150.0 g
Energy:			
Total calories, approx.	700 kcal	1050 kcal	1400 kcal
Non-protein calories	600 kcal	900 kcal	1200 kcal
Glucose calories	300 kcal	450 kcal	600 kcal
Lipid calories ^a	300 kcal	450 kcal	600 kcal
Non-protein calories / nitrogen ratio	150 kcal/g	150 kcal/g	150 kcal/g
Glucose / lipid calories ratio	50/50	50/50	50/50
Lipid / total calories	43%	43%	43%
Electrolytes:			
Sodium	21.0 mmol	31.5 mmol	42.0 mmol
Potassium	16.0 mmol	24.0 mmol	32.0 mmol
Magnesium	2.2 mmol	3.3 mmol	4.4 mmol
Calcium	2.0 mmol	3.0 mmol	4.0 mmol
Phosphate ^b	8.5 mmol	12.7 mmol	17.0 mmol
Acetate	27 mmol	41 mmol	55 mmol
Chloride	24 mmol	37 mmol	49 mmol
pH	6.4	6.4	6.4
Osmolarity	760 mosm/l	760 mosm/l	760 mosm/l

a: Includes calories from purified egg phospholipids
b: Includes phosphate provided by the lipid emulsion

3. PHARMACEUTICAL FORM

After reconstitution: Emulsion for infusion.

Appearance prior to reconstitution:

- The amino acids and glucose solutions are clear, colourless or slightly yellow.
- The lipid emulsion is homogenous with a milky appearance.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

PeriOLIMEL N4E is indicated for parenteral nutrition for adults and children greater than 2 years of age when oral or enteral nutrition is impossible, insufficient or contraindicated.

4.2 Posology and method of administration

Posology

PeriOLIMEL N4E is not recommended for use in children less than 2 years of age due to inadequate composition and volume (see sections 4.4; 5.1 and 5.2).

The maximum daily dose mentioned below should not be exceeded. Due to the static composition of the multi-chamber bag, the ability to simultaneously meet all nutrient needs of the patient may not be possible. Clinical situations may exist where patients require amounts of nutrients varying from the composition of the static bag. In this situation any volume (dose) adjustments must take into consideration the resultant effect this will have on the dosing of all other nutrient components of PeriOLIMEL N4E.

In adults

The dosage depends on the patient’s energy expenditure, clinical status, body weight, and the ability to metabolise the constituents of PeriOLIMEL N4E, as well as additional energy or proteins provided orally/enterally; therefore, the bag size should be chosen accordingly.

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The average daily requirements are:

- 0.16 to 0.35 g nitrogen/kg body weight (1 to 2 g of amino acids/kg), depending on the patient's nutritional status and degree of catabolic stress,
- 20 to 40 kcal/kg,
- 20 to 40 ml fluid /kg, or 1 to 1.5 ml per expended kcal.

For PeriOLIMEL N4E, the maximal daily dose is defined by fluid intake, 40 ml/kg, corresponding to 1 g/kg amino acids, 3 g/kg glucose, 1.2 g/kg lipids, 0.8 mmol/kg sodium, and 0.6 mmol/kg potassium. For a 70kg patient, this would be equivalent to 2800 ml PeriOLIMEL N4E per day, resulting in an intake of 71 g amino acids, 210 g glucose and 84 g lipids (i.e. 1680 non-protein kcal and 1960 total kcal).

Normally, the flow rate must be increased gradually during the first hour and then be adjusted to take into account the dose being administered, the daily volume intake and the duration of the infusion.

For PeriOLIMEL N4E, the maximal infusion rate is 3.2 ml/kg/hour, corresponding to 0.08 g/kg/hour amino acids, 0.24 g/kg/hour glucose and 0.10 g/kg/hour lipids.

In children greater than 2 years of age and adolescents

There have been no studies performed in the paediatric population.

The dosage depends on the patient's energy expenditure, clinical status, body weight, and the ability to metabolise constituents of PeriOLIMEL N4E, as well as additional energy or proteins given orally/enterally; therefore, the bag size should be chosen accordingly.

In addition, daily fluid, nitrogen, and energy requirements continuously decrease with age. Two groups, ages 2 to 11 years and 12 to 18 years, are considered.

For PeriOLIMEL N4E, in both age groups, the magnesium concentration is the limiting factor for daily dose. In the 2 to 11 year age group, the lipid concentration is the limiting factor for hourly rate. In the 12 to 18 year age group, the glucose concentration is the limiting factor for hourly rate. The resulting intakes are displayed below:

Constituent	2 to 11 years		12 to 18 years	
	Recommended ^a	PeriOLIMEL N4E Max Vol	Recommended ^a	PeriOLIMEL N4E Max Vol
Maximum Daily Dose				
Fluids (ml/kg/d)	60 – 120	45	50 – 80	45
Amino acids (g/kg/d)	1 – 2 (up to 2.5)	1.1	1 – 2	1.1
Glucose (g/kg/d)	1.4 – 8.6	3.4	0.7 – 5.8	3.4
Lipids (g/kg/d)	0.5 to 3	1.4	0.5 to 2 (up to 3)	1.4
Total energy (kcal/kg/d)	30 – 75	31.5	20 – 55	31.5
Maximum Hourly Rate				
PeriOLIMEL N4E (ml/kg/h)		4.3		3.2
Amino acids (g/kg/h)	0.20	0.11	0.12	0.08
Glucose (g/kg/h)	0.36	0.33	0.24	0.24
Lipids (g/kg/h)	0.13	0.13	0.13	0.10

^a Recommended values from 2018 ESPGHAN/ESPEN/ESPR Guidelines

Normally, the flow rate must be increased gradually during the first hour and then be adjusted to take into account the dose being administered, the daily volume intake, and the duration of the infusion.

In general, it is recommended to start the infusion for small children with low daily dose and gradually increase it up to the maximal dosage (see above).

Method and duration of administration

For single use only.

It is recommended that, after opening the bag, the contents are used immediately and not stored for subsequent infusion.

After reconstitution, the mixture is homogenous with a milky appearance.

For instructions for preparation and handling of the emulsion for infusion see section 6.6.

Due to its low osmolarity, PeriOLIMEL N4E can be administered through a peripheral or central vein. The recommended duration of infusion for a parenteral nutrition bag is between 12 and 24 hours.

Treatment with parenteral nutrition may be continued for as long as required by the patient's clinical conditions.

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4.3 Contraindications

The use of PeriOLIMEL N4E is contraindicated in the following situations:

- In premature neonates, infants and children less than 2 years of age,
- Hypersensitivity to egg, soya-bean, peanut proteins, or corn/corn products (see section 4.4) or to any of the active substances or excipients, listed in section 6.1,
- Congenital abnormalities of amino acid metabolism,
- Severe hyperlipidemia or severe disorders of lipid metabolism characterised by hypertriglyceridaemia,
- Severe hyperglycaemia,
- Pathologically-elevated plasma concentration of sodium, potassium, magnesium, calcium and/or phosphorus.

4.4 Special warnings and precautions for use

An excessively fast administration of total parenteral nutrition (TPN) solutions may result in severe or fatal consequences.

The infusion must be stopped immediately if any signs or symptoms of an allergic reaction (such as sweating, fever, chills, headache, skin rashes or dyspnea) develop. This medicinal product contains soya-bean oil, and egg phospholipids. Soya-bean and egg proteins may cause hypersensitivity reactions. Cross-allergic reactions between soya-bean and peanut proteins have been observed.

PeriOLIMEL N4E contains glucose derived from corn, which may cause hypersensitivity reactions in patients with allergy to corn or corn products (see section 4.3).

Ceftriaxone must not be mixed or administered simultaneously with any calcium-containing IV solutions even via different infusion lines or different infusion sites. Ceftriaxone and calcium-containing solutions may be administered sequentially one after another if infusion lines at different sites are used or if the infusion lines are replaced or thoroughly flushed between infusions with physiological salt-solution to avoid precipitation. In patients requiring continuous infusion with calcium-containing TPN solutions, healthcare professionals may wish to consider the use of alternative antibacterial treatments which do not carry a similar risk of precipitation. If use of ceftriaxone is considered necessary in patients requiring continuous nutrition, TPN solutions and ceftriaxone can be administered simultaneously, albeit via different infusion lines at different sites. Alternatively, infusion of TPN solution could be stopped for the period of ceftriaxone infusion, considering the advice to flush infusion lines between solutions (see sections 4.5 and 6.2).

Pulmonary vascular precipitates causing pulmonary vascular embolism and respiratory distress have been reported in patients receiving parenteral nutrition. In some cases, fatal outcomes have occurred. Excessive addition of calcium and phosphate increases the risk of formation of calcium phosphate precipitates (see section 6.2).

Suspected precipitate formation in the blood stream has also been reported.

In addition to inspection of the solution, the infusion set and catheter should also periodically be checked for precipitates.

If signs of respiratory distress occur, the infusion should be stopped and medical evaluation initiated.

Do not add other medicinal products or substances to any components of the bag or to the reconstituted emulsion without first confirming their compatibility and the stability of the resulting preparation (in particular, the stability of the lipid emulsion).

Formation of precipitates or destabilization of the lipid emulsion could result in vascular occlusion (see section 6.2 and 6.6).

Severe water and electrolyte equilibration disorders, severe fluid overload states, and severe metabolic disorders must be corrected before starting the infusion.

Specific clinical monitoring is required when an intravenous infusion is started.

Vascular-access infection and sepsis are complications that may occur in patients receiving parenteral nutrition, particularly in case of poor maintenance of catheters, immunosuppressive effects of illness or drugs. Careful monitoring of signs, symptoms and laboratory test results for fever/chills, leukocytosis, technical complications with the access device, and hyperglycemia can help recognize early infections. Patients who require parenteral nutrition are often predisposed to infectious complications due to malnutrition and/or their underlying disease state. The occurrence of septic complications can be decreased with heightened emphasis on aseptic techniques in catheter placement and maintenance, as well as aseptic techniques in the preparation of the nutritional formula.

Monitor water and electrolyte balance, serum osmolarity, serum triglycerides, acid/base balance, blood glucose, liver and kidney function tests, coagulation tests, and blood count, including platelets, throughout treatment.

Elevated liver enzymes and cholestasis have been reported with similar products. Monitoring of serum ammonia should be considered if hepatic insufficiency is suspected.

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Metabolic complications may occur if the nutrient intake is not adapted to the patient's requirements, or the metabolic capacity of any given dietary component is not accurately assessed. Adverse metabolic effects may arise from administration of inadequate or excessive nutrients or from inappropriate composition of an admixture for a particular patient's needs.

Administration of amino acid solutions may precipitate acute folate deficiency; folic acid is, therefore, recommended to be given daily.

Extravasation

Catheter site should be monitored regularly to identify signs of extravasation. If extravasation occurs the administration should be stopped immediately, keeping the inserted catheter or cannula in place for immediate management of the patient. If possible, aspiration should be performed through the inserted catheter/ cannula in order to reduce the amount of fluid present in the tissues before removing the catheter/ cannula. When involving an extremity, the concerned limb should be elevated.

Depending on the extravasated product (including the product(s) being mixed with PeriOLIMEL N4E, if applicable) and the stage/extent of any injury, appropriate specific measures should be taken. Options for management may include non-pharmacologic, pharmacologic and/or surgical intervention. In case of large extravasation, plastic surgeon advice should be sought within the first 72 hours. The extravasation site should be monitored at least every 4 hours during the first 24 hours, then once daily. The infusion should not be restarted in the same peripheral or central vein.

Hepatic insufficiency

Use with caution in patients with hepatic insufficiency because of the risk of developing or worsening neurological disorders associated with hyperammonaemia. Regular clinical and laboratory tests are required, particularly liver function parameters, blood glucose, electrolytes and triglycerides.

Renal insufficiency

Use with caution in patients with renal insufficiency, particularly if hyperkalaemia is present, because of the risk of developing or worsening metabolic acidosis and hyperazotemia if extra-renal waste removal is not being performed. Fluid, triglycerides and electrolytes status should be closely monitored in these patients.

Hematologic

Use with caution in patients with coagulation disorders and anaemia. Blood count and coagulation parameters should be closely monitored.

Endocrine and metabolism

- Use with caution in patients with:
- Metabolic acidosis. The administration of carbohydrates is not recommended in the presence of lactic acidosis. Regular clinical and laboratory tests are required.
 - Diabetes mellitus. Monitor glucose concentrations glucosuria, ketonuria and, where applicable adjust insulin dosages.
 - Hyperlipidaemia due to the presence of lipids in the emulsion for infusion. Regular clinical and laboratory tests are required.
 - Amino acid metabolism disorders

Hepatobiliary disorders

Hepatobiliary disorders including cholestasis, hepatic steatosis, fibrosis and cirrhosis, possibly leading to hepatic failure, as well as cholecystitis and cholelithiasis are known to develop in some patients on parenteral nutrition. The etiology of these disorders is thought to be multifactorial and may differ between patients. Patients developing abnormal laboratory parameters or other signs of hepatobiliary disorders should be assessed early by a clinician knowledgeable in liver diseases in order to identify possible causative and contributory factors, and possible therapeutic and prophylactic interventions.

Serum triglycerides concentrations and the ability of the body to remove lipids must be checked regularly.

Serum triglycerides concentrations must not exceed 3 mmol/l during the infusion.

If a lipid metabolism abnormality is suspected, it is recommended to measure daily serum triglycerides levels after a period of 5 to 6 hours without administering lipids. In adults, the serum must be clear in less than 6 hours after stopping the infusion containing the lipid emulsion. The next infusion must only be administered when the serum triglycerides concentrations have returned to baseline values.

Fat overload syndrome has been reported with similar products. The reduced or limited ability to metabolise the lipids contained in PeriOLIMEL N4E may result in a "fat overload syndrome" which may be caused by overdose; however, the signs and symptoms of this syndrome may also occur when the product is administered according to instructions (see also section 4.8).

In the event of hyperglycaemia, the infusion rate of PeriOLIMEL N4E must be adjusted and/or insulin administered.

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Thrombophlebitis may develop if peripheral veins are used. The catheter insertion site must be monitored daily for local signs of thrombophlebitis.

When making additions, the final osmolarity of the mixture must be measured before administration. The mixture obtained must be administered through a central or peripheral venous line depending on its final osmolarity. If the final mixture administered is hypertonic, it may cause irritation of the vein when administered into a peripheral vein.

Although there is a natural content of trace elements and vitamins in the product, the levels are insufficient to meet body requirements. Trace elements and vitamins, should be added in sufficient quantities to meet individual patient requirements and to prevent deficiencies from developing. See instructions for making additions to this product.

Caution should be exercised in administering PeriOLIMEL N4E to patients with increased osmolarity, adrenal insufficiency, heart failure or pulmonary dysfunction.

In malnourished patients, initiation of parenteral nutrition can precipitate fluid shifts resulting in pulmonary oedema and congestive heart failure, as well as a decrease in the serum concentration of potassium, phosphorus, magnesium or water-soluble vitamins. These changes can occur within 24 to 48 hours; therefore, careful and slow initiation of parenteral nutrition is recommended together with close monitoring and appropriate adjustments of fluid, electrolytes, trace elements and vitamins.

Do not connect bags in series in order to avoid the possibility of air embolism due to residual gas contained in the primary bag.

To avoid risks associated with excessively rapid infusion rates, it is recommended to use a continuous and controlled infusion.

PeriOLIMEL N4E must be administered with caution to patients with a tendency towards electrolyte retention.

Intravenous infusion of amino acids is accompanied by increased urinary excretion of trace elements, in particular copper and zinc. This should be taken into account in the dosing of trace elements, especially during long-term intravenous nutrition.

Interference with laboratory tests

The lipids contained in this emulsion may interfere with the results of certain laboratory tests (see section 4.5).

Special precautions in paediatrics

When administered to children greater than 2 years of age, it is essential to use a bag that has a volume corresponding to the daily dosage.

- PeriOLIMEL N4E is not suitable for use in children less than 2 years of age because:
- The glucose intake is too low, leading to a low glucose / lipid ratio,
 - The absence of cysteine makes the amino acids profile inadequate,
 - Calcium is too low,
 - The bag volumes are not appropriate.

Maximal infusion rate is 4.3 ml/kg/hour in children 2 to 11 years of age and 3.2 mL/kg/hour in children 12 to 18 years of age.

Vitamin and trace elements supplementation is always required. Paediatric formulations must be used.

Geriatric population

In general, dose selection for an elderly patient should be cautious, reflecting the greater frequency of decreased hepatic, renal, or cardiac function, and of concomitant disease or other drug therapy.

4.5 Interaction with other medicinal products and other forms of interaction

No interaction studies have been performed.

PeriOLIMEL N4E must not be administered simultaneously with blood through the same infusion tubing because of the possibility of pseudoagglutination.

The lipids contained in this emulsion may interfere with the results of certain laboratory tests (for example, bilirubin, lactate dehydrogenase, oxygen saturation, blood haemoglobin) if the blood sample is taken before the lipids are eliminated (these are generally eliminated after a period of 5 to 6 hours without receiving lipids).

Precipitation of ceftriaxone-calcium can occur when ceftriaxone is mixed with calcium-containing solutions in the same intravenous administration line. Ceftriaxone must not be mixed or administered simultaneously with calcium-containing intravenous solutions, including PeriOLIMEL N4E, through the same infusion line (e.g., via Y-site). However, ceftriaxone and calcium-containing solutions may be administered sequentially of one another if the infusion lines are thoroughly flushed between infusions with a compatible fluid (see sections 4.4 and 6.2).

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PeriOLIMEL N4E contains vitamin K, naturally present in lipid emulsions. The amount of vitamin K in recommended doses of PeriOLIMEL N4E are not expected to influence effects of coumarin derivatives.

Due to the potassium content of PeriOLIMEL N4E, special care should be taken in patients treated with potassium-sparing diuretics (e.g. amiloride spironolactone, triamterene), angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor antagonists, or the immunosuppressants tacrolimus or cyclosporine in view of the risk of hyperkalemia.

Some medicinal products, like insulin, may interfere with the body's lipase system. This kind of interaction seems, however, to be of limited clinical importance.

Heparin given in clinical doses causes a transient release of lipoprotein lipase into the circulation. This may result initially in increased plasma lipolysis followed by a transient decrease in triglyceride clearance.

4.6 Fertility, pregnancy and lactation

Pregnancy

There are no clinical data from the use of PeriOLIMEL N4E in pregnant women. No animal reproductive studies have been performed with PeriOLIMEL N4E (see section 5.3). Taking into account the use and indications of PeriOLIMEL N4E, the product may be considered during pregnancy, if necessary. PeriOLIMEL N4E should only be given to pregnant women after careful consideration.

Breast-feeding

There is insufficient information on the excretion of PeriOLIMEL N4E components/metabolites in human milk. Parenteral nutrition may become necessary during breast-feeding. PeriOLIMEL N4E should only be given to breast-feeding women after careful consideration.

Fertility

No adequate data are available.

4.7 Effects on ability to drive and use machines

Not relevant.

4.8 Undesirable effects

Potential undesirable effects may occur as a result of inappropriate use (for example: overdose, excessively fast infusion rate) (see sections 4.4 and 4.9).

At the beginning of the infusion, any of the following abnormal signs (sweating, fever, shivering, headache, skin rashes, dyspnoea) should be cause for immediate discontinuation of the infusion.

The adverse drug reactions (ADRs) reported with OLIMEL N9-840 in a randomized, double-blind, active-controlled efficacy and safety study, are listed in the table below. Twenty-eight patients with various medical conditions (i.e., postsurgical fasting, severe malnutrition, enteral intake insufficient or forbidden) were included and treated; patients in the OLIMEL group received drug product up to 40 ml/kg/day over 5 days.

The pooled data from clinical trials and the postmarketing experience indicate the following adverse drug reactions (ADRs) related to PeriOLIMEL N4E.

System Organ Class	MedDRA Preferred Term	Frequency ^a
Immune System Disorders	Hypersensitivity reactions including hyperhidrosis, pyrexia, chills, headache, skin rash (erythematous, papular, pustular, macular, generalised rash), pruritus, hot flush, dyspnoea	Not known ^b
Cardiac disorders	Tachycardia	Common ^a
Metabolism and nutrition disorders	Decreased appetite	Common ^a
	Hypertriglyceridaemia	Common ^a
Gastrointestinal disorders	Abdominal pain	Common ^a
	Diarrhoea	Common ^a
	Nausea	Common ^a
	Vomiting	Not known ^b
Vascular disorders	Hypertension	Common ^a
General disorders and administration site conditions	Extravasation which may result at infusion site level in: pain, irritation, swelling/oedema, erythema/warmth, skin necrosis, blisters/vesicles, inflammation, induration, skin tightness	Not known ^b

^a: Frequency is defined as very common (≥1/10); common (≥1/100, <1/10); uncommon (≥1/1,000, <1/100); rare (≥1/10,000, <1/1,000); very rare (<1/10,000); and not known (cannot be estimated from the available data).

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- b: ADRs reported during post-marketing experience with PeriOLIMEL N4E
- The following class-like-adverse drug reactions (ADRs) have been described in other sources in relation to similar parenteral nutrition products; the frequency of these events is not known.
- Blood and lymphatic system disorders: thrombocytopenia
 - Hepatobiliary disorders: cholestasis, hepatomegaly, jaundice
 - Immune system disorders: hypersensitivity
 - Injury, poisoning and procedural complications: Parenteral nutrition associated liver disease (see section 4.4, sub-section "Hepatobiliary disorders")
 - Investigations: blood alkaline phosphatase increased, transaminases increased, blood bilirubin increased, elevated liver enzymes
 - Renal and urinary disorders: azotemia
 - Vascular disorders: Pulmonary vascular precipitates (pulmonary vascular embolism and respiratory distress) (see section 4.4)

Fat overload syndrome (very rare)

Fat overload syndrome has been reported with similar products. This may be caused by inappropriate administration (e.g. overdose and/or infusion rate higher than recommended, see section 4.9); however, the signs and symptoms of this syndrome may also occur at the start of an infusion when the product is administered according to instructions. The reduced or limited ability to metabolize the lipids contained in PeriOLIMEL N4E accompanied by prolonged plasma clearance may result in a "fat overload syndrome". This syndrome is associated with a sudden deterioration in the patient's clinical condition and is characterized by findings such as fever, anemia, leukopenia, thrombocytopenia, coagulation disorders, hyperlipidemia, liver fatty infiltration (hepatomegaly), deteriorating liver function, and central nervous system manifestations (e.g. coma). The syndrome is usually reversible when infusion of the lipid emulsion is stopped.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions.

4.9 Overdose

In the event of inappropriate administration (overdose and/or infusion rate higher than recommended), signs of hypervolaemia and acidosis may occur.

An excessively fast infusion or administration of an inappropriately large volume of the product may cause nausea, vomiting, chills, headache, hot flush, hyperhidrosis and electrolyte disturbances. In such situations the infusion must be stopped immediately.

Hyperglycaemia, glucosuria, and a hyperosmolar syndrome may develop if the glucose infusion rate exceeds clearance.

The reduced or limited ability to metabolise lipids may result in a "fat overload syndrome", the results of which are usually reversible after the infusion of the lipid emulsion is stopped (see also section 4.8).

In some serious cases, haemodialysis, haemofiltration or haemodiafiltration may be necessary.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Solutions for parenteral nutrition/ combinations

ATC code: B05BA10.

PeriOLIMEL N4E's content in nitrogen (L series amino acids) and energy (glucose and triglycerides) enables maintaining an adequate nitrogen/energy balance.

This formulation also contains electrolytes.

The lipid emulsion included in PeriOLIMEL N4E is an association of refined olive oil and refined soya-bean oil (ratio 80/20), with the following approximate distribution of fatty acids:

- 15% saturated fatty acids (SFA)
- 65% monounsaturated fatty acids (MUFA)
- 20% polyunsaturated essential fatty acids (PUFA)

The phospholipid/triglyceride ratio is 0.06.

Olive oil contains significant amount of alpha-tocopherol which, combined with a moderate PUFA intake, contribute to improved vitamin E status and the reduction of lipid peroxidation.

The amino acids solution contains 17 L-series amino acids (including 8 essential amino acids), which are required for protein synthesis.

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Amino acids also represent an energy source. Their oxidation results in excretion of nitrogen in the form of urea.

The amino acids profile is as follows:

- Essential amino acids/total amino acids: 44.8%
- Essential amino acids (g)/total nitrogen (g): 2.8%
- Branched-chain amino acids/total amino acids: 18.3%

The carbohydrate source is glucose.

CLINICAL TRIAL

Study ICS1063B/P01/03/Mu.F was a prospective randomised double-blind multicenter study performed in fifty six hospitalised patients (age range 18-85 years) to evaluate safety and nutritional efficacy of OLIMEL N9-840 compared to OliClinomel N8-800. The study was conducted in a variety of patients (primarily post-surgery and trauma) who required balanced parenteral nutrition representing at least 50% of the daily nonprotein energy requirements for 5 days. The primary nutritional efficacy endpoint was transthyretin (pre-albumin) levels. Safety was evaluated using adverse events, vital signs, and biochemical markers for renal (urea, creatinine), hepatic (AST, ALT, alkaline phosphatase, GGT, bilirubin), hematologic (RBC count, hemoglobin, hematocrit, platelets, WBCs, lymphocytes, neutrophils, monocytes, eosinophils, basophils), organ functions as well as glucose and lipid parameters (triglycerides, cholesterol).

Efficacy analysis on the per protocol (PP) and intent-to-treat (ITT) populations showed no difference between the OLIMEL and OliClinomel groups on the primary endpoint (transthyretin), which improved from baseline to Day 5/end of treatment.

Changes in Mean Prealbumin (Transthyretin) Levels – Study

ICS1063B/P01/03/Mu.F

Treatment Group Study Population	Mean ± SD Transthyretin Levels (g/L)	
	Baseline	Day 5/EOT
OLIMEL N9-840		
Intent-to-Treat Population (n = 24)	0.144 ± 0.075	0.206 ± 0.142
Per Protocol Population (n = 24)	0.144 ± 0.075	0.206 ± 0.142
OliClinomel N8-800		
Intent-to-Treat Population (n = 26)	0.146 ± 0.083	0.181 ± 0.082
Per Protocol Population (n = 23)	0.139 ± 0.078	0.172 ± 0.080

EOT: End of treatment. SD: Standard deviation.

The safety of the two formulations was comparable. There was no difference between the treatment groups for any clinical laboratory or vital sign parameters evaluated during the study.

5.2 Pharmacokinetic properties

The ingredients of PeriOLIMEL N4E (amino acids, electrolytes, glucose and lipids) are distributed, metabolised and removed in the same way as if they had been administered individually.

The pharmacokinetic properties of the amino acids administered intravenously are principally the same as those of amino acids supplied by oral feeding. Amino acids from food proteins, however, first pass through the vena porta before reaching the systemic circulation.

The elimination rate of lipid emulsions depends on particle size. Small lipid particles appear to delay clearance whereas they increase lipolysis by lipoprotein lipase.

The size of the lipid particles in the emulsion is close to that of chylomicrons and this emulsion therefore has a similar elimination rate.

5.3 Preclinical safety data

No preclinical studies with PeriOLIMEL N4E have been performed.

Preclinical toxicity studies performed using the lipid emulsion contained in PeriOLIMEL N4E have identified the changes, which are conventionally found with a high intake of a lipid emulsion: fatty liver, thrombocytopaenia and elevated cholesterol.

Preclinical studies performed using the solutions of amino acids and glucose contained in PeriOLIMEL N4E of different qualitative compositions and concentrations have not, however, revealed any specific toxicity.

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6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Lipid emulsion compartment:

Purified egg phospholipids, Glycerol, Sodium oleate, Sodium hydroxide (for pH adjustment), Water for injections.

Compartment of amino-acid solution with electrolytes:

Glacial acetic acid (for pH adjustment), Water for injections.

Compartment of glucose solution with calcium:

Hydrochloric acid (for pH adjustment), Water for injections.

6.2 Incompatibilities

Do not add other medicinal products or substances to any components of the bag or to the reconstituted emulsion without first confirming their compatibility and the stability of the resulting preparation (in particular, the stability of the lipid emulsion).

Incompatibilities may be produced for example, by excessive acidity (low pH) or inappropriate content of divalent cations (Ca^{2+} and Mg^{2+}), which may destabilize the lipid emulsion.

As with any parenteral nutrition admixture, calcium and phosphate ratios must be considered. Excess addition of calcium and phosphate, especially in the form of mineral salts, may result in the formation of calcium phosphate precipitates.

PeriOLIMEL N4E contains calcium ions which pose additional risk of coagulation precipitated in citrate anticoagulated/preserved blood or components.

Ceftriaxone must not be mixed or administered simultaneously with intravenous calcium-containing solutions, including PeriOLIMEL N4E, through the same infusion line (e.g., via Y-connector) because of the risk of precipitation of ceftriaxone-calcium salt (see section 4.4 and 4.5).

Due to the risk of precipitation, PeriOLIMEL N4E should not be administered through the same infusion line or admixed together with ampicillin or fosphenytoin.

Check compatibility with solutions administered simultaneously through the same administration set, catheter or cannula.

Do not administer before, simultaneously with or after blood through the same equipment because of the risk of pseudoagglutination.

6.3 Shelf life

2 years if the overwrap is not damaged.

After reconstitution

It is recommended that the product be used immediately after the non-permanent seals between the 3 compartments have been opened. However, the stability of the reconstituted emulsion has been demonstrated for 7 days (between 2°C and 8°C) followed by 48 hours at temperature not exceeding 30°C.

After addition of supplements (electrolytes, trace elements and vitamins; see section 6.6)

For specific admixtures, in-use stability has been demonstrated for 7 days (between 2°C and 8°C) followed by 48 hours at temperature not exceeding 25°C.

From a microbiological point of view, any admixture should be used immediately. If not used immediately, storage times and conditions, after mixing and prior to use, are the responsibility of the user and would normally not be longer than 24 hours at 2°C to 8°C, unless addition of supplements has taken place in controlled and validated aseptic conditions.

6.4 Special precautions for storage

Do not freeze.

Store in the overpouch.

Store at temperature not exceeding 30°C.

For storage conditions of the reconstituted medicinal product, see section 6.3.

6.5 Nature and contents of container

The 3-compartment bag is a multilayer plastic bag. The inner (contact) layer of the bag material is made of a blend of polyolefinic copolymers and is compatible with amino acid solutions, glucose solutions, and lipid emulsions. Other layers are made of polyethylene vinyl acetate (EVA), and of copolyester.

The glucose compartment is fitted with an injection site to be used for addition of supplements.

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The amino acids compartment is fitted with an administration site for insertion of the spike of the infusion set.
The bag is packaged in an oxygen barrier overpouch with an oxygen absorber sachet.

Pack sizes:
1000 ml bag: 1 carton with 6 bags
1500 ml bag: 1 carton with 4 bags
2000 ml bag: 1 carton with 4 bags
1 bag of 1000 ml, 1500 ml and 2000 ml

Not all pack sizes may be marketed.

6.6 Special precautions for disposal and other handling

To open

Remove the protective overpouch.
Discard the oxygen absorber sachet.

Confirm the integrity of the bag and of the nonpermanent seals. Use only if the bag is not damaged, if the nonpermanent seals are intact (i.e. no mixture of the contents of the 3 compartments), if the amino acids solution and the glucose solution are clear, colourless, or slightly yellow, and practically free of visible particles; and if the lipid emulsion is a homogeneous liquid with a milky appearance.

Mixing the solutions and the emulsion

Ensure that the product is at room temperature when breaking the nonpermanent seals.
Manually roll the bag onto itself, starting at the top of the bag (hanger end). The nonpermanent seals will disappear from the side near the inlets. Continue to roll the bag until the seals are open along approximately half of their length.
Mix by inverting the bag at least 3 times.
After reconstitution, the mixture is a homogeneous emulsion with a milky appearance.

Additions

The capacity of the bag is sufficient to enable additions such as vitamins, electrolytes and trace elements.
Any addition (including vitamins) may be made into the reconstituted mixture (after the nonpermanent seals have been opened and after the contents of the 3 compartments have been mixed).
Vitamins may also be added into the glucose compartment before the mixture is reconstituted (before opening the nonpermanent seals and before mixing the 3 compartments).
When making additions to formulations containing electrolytes, the amount of electrolytes already present in the bag should be taken into account.
Additions must be performed by qualified personnel under aseptic conditions.
PeriOLIMEL N4E may be supplemented with electrolytes according to the tables below:

Per 1000 ml			
	Included level	Maximal further addition	Maximal total level
Sodium	21 mmol	129 mmol	150 mmol
Potassium	16 mmol	134 mmol	150 mmol
Magnesium	2.2 mmol	3.4 mmol	5.6 mmol
Calcium	2.0 mmol	3.0 (1.5 ^a) mmol	5.0 (3.5 ^a) mmol
Inorganic Phosphate	0 mmol	8.0 mmol	8.0 mmol
Organic Phosphate	8.5 mmol ^b	15.0 mmol	23.5 mmol ^b

a: Value corresponding to the addition of inorganic phosphate.
b: Including phosphate provided by the lipid emulsion

Trace elements and vitamins:

Stability has been demonstrated with commercially-available preparations of vitamins and trace elements (containing up to 1 mg of iron).
Compatibility for other additives is available upon request.
When making additions, the final osmolarity of the mixture must be measured before administration via a peripheral vein.
To perform an addition:
• Aseptic conditions must be observed.
• Prepare the injection site of the bag.

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- Puncture the injection site and inject the additives using an injection needle or a reconstitution device.
- Mix content of the bag and the additives.

Preparation of the infusion

Aseptic conditions must be observed. Suspend the bag.

Remove the plastic protector from the administration outlet.

Firmly insert the spike of the infusion set into the administration outlet.

Administration

For single use only

Only administer the product after the nonpermanent seals between the 3 compartments have been broken and the contents of the 3 compartments have been mixed.

Ensure that the final emulsion for infusion does not show any evidence of phase separation.

After opening the bag, the contents must be used immediately. The opened bag must never be stored for a subsequent infusion. Do not reconnect any partially used-bag.

Do not connect bags in series in order to avoid the possibility of air embolism due to gas contained in the primary bag.

Any unused product or waste material and all necessary devices must be discarded.

7. MANUFACTURER

Baxter S.A.,
Boulevard Rene Branquart, 80
7860 Lessines, Belgium.

DATE OF REVISION

September 2023